

5-bit R-2R DAC

The Electric VLSI Design System program was used to develop and verify the function of a 5-bit R-2R ladder DAC in conjunction with LTspice XVII. In the design, R has a resistance of 10 k Ω and the resistors are n-well type. In the implementation of each 2R branch, two resistors are used in series. The output reached when the word b3b2b1b0b1 is applied to the circuit is expressed as follows:

$$V_{out} = V_{DD} (b_4/2 + b_3/4 + b_2/8 + b_1/16 + b_0/32)$$

With $V_{DD} = 5$ V,

$$1 \text{ LSB} = 5/32 = 0.15625 \text{ V}, \quad V_{FS} = (31/32)(5) = 4.84375 \text{ V}$$

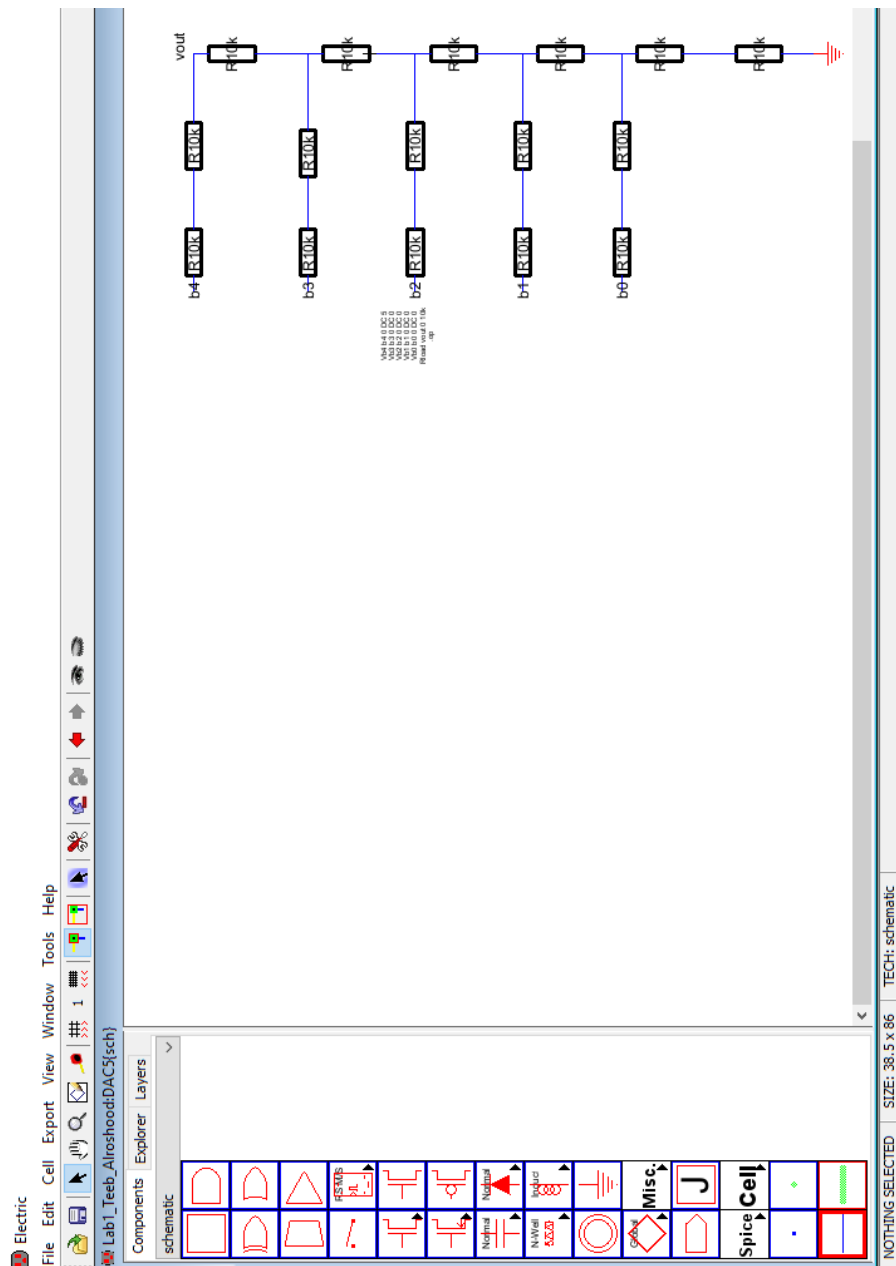


Figure 1. Electric schematic of the 5-bit R-2R DAC. Each 2R branch uses two separate 10 k Ω resistors.

Output resistance

When finding the output resistance, it is required that all voltage sources be turned down to a reference value which is zero. The process proceeds up the ladder where every level of the configuration introduces the same impedance,

$$2R \parallel 2R = R$$

We can repeat the same procedure to get the upper levels as well:

$$2R \parallel (R + R) = 2R \parallel 2R = R$$

Thus, the final result in case of output resistance is also $R = 10 \text{ k}\Omega$.

Functional verification

The LTspice operating-point results agree with the ideal DAC equation. The measured values used in the report are listed below.

Input	Expected V _{out}	LTspice V _{out}
00000	0 V	0 V
00001	0.15625 V	0.15625 V
10000	2.50000 V	2.50000 V
11000	3.75000 V	3.75000 V
10101	3.28125 V	3.28125 V
11111	4.84375 V	4.84375 V

LTspice XVII - [DAC5.spi]

File Edit View Simulate Tools Window Help

DAC5.spi

```
.SUBCKT Lab1_Teeb_Alroshood_R10k in out
Rresnwell101 out in 10k
.ENDS Lab1_Teeb_Alroshood_R10k

.global gnd

*** TOP LEVEL CELL: DAC5{sch}
XR10k00 b4 net@0 Lab1_Teeb_Alroshood_R10k
XR10k01 net@0 vout Lab1_Teeb_Alroshood_R10k
XR10k03 vout net@15 Lab1_Teeb_Alroshood_R10k
XR10k04 b3 net@3 Lab1_Teeb_Alroshood_R10k
XR10k05 net@3 net@15 Lab1_Teeb_Alroshood_R10k
XR10k06 net@15 net@16 Lab1_Teeb_Alroshood_R10k
XR10k07 b2 net@6 Lab1_Teeb_Alroshood_R10k
XR10k08 net@6 net@16 Lab1_Teeb_Alroshood_R10k
XR10k09 net@16 net@17 Lab1_Teeb_Alroshood_R10k
XR10k10 b1 net@9 Lab1_Teeb_Alroshood_R10k
XR10k11 net@9 net@17 Lab1_Teeb_Alroshood_R10k
XR10k12 net@17 net@18 Lab1_Teeb_Alroshood_R10k
XR10k13 b0 net@12 Lab1_Teeb_Alroshood_R10k
XR10k14 net@12 net@18 Lab1_Teeb_Alroshood_R10k
XR10k15 net@18 net@21 Lab1_Teeb_Alroshood_R10k
XR10k16 net@21 gnd Lab1_Teeb_Alroshood_R10k

* Spice Code nodes in cell cell 'DAC5{sch}'
Vb4 b4 0 DC 0
Vb3 b3 0 DC 0
Vb2 b2 0 DC 0
Vb1 b1 0 DC 0
Vb0 b0 0 DC 0
.op
.END
```

Ready

*** SPICE deck for cell DAC5{sch} from library Lab1_Teeb_Alroshood

```
--- Operating Point ---
V(net@0): 0 voltage
V(b4): 0 voltage
V(vout): 0 voltage
V(net@15): 0 voltage
V(net@3): 0 voltage
V(b3): 0 voltage
V(net@16): 0 voltage
V(net@6): 0 voltage
V(b2): 0 voltage
V(net@17): 0 voltage
V(net@9): 0 voltage
V(b1): 0 voltage
V(net@18): 0 voltage
V(net@12): 0 voltage
V(b0): 0 voltage
V(net@21): 0 voltage
I(Vb0): 0 device_current
I(Vb1): 0 device_current
I(Vb2): 0 device_current
I(Vb3): 0 device_current
I(Vb4): 0 device_current
Ix(r10k00:IN): 0 subckt_current
Ix(r10k00:OUT): 0 subckt_current
Ix(r10k01:IN): 0 subckt_current
Ix(r10k01:OUT): 0 subckt_current
Ix(r10k03:IN): 0 subckt_current
Ix(r10k03:OUT): 0 subckt_current
Ix(r10k04:IN): 0 subckt_current
```

Input 00000: $V_{out} = 0\text{ V}$.

LTspice XVII - [DAC5.spi]

File Edit View Simulate Tools Window Help

DAC5.spi

```
.SUBCKT Lab1_Teeb_Alroshood_R10k in out
Rresnwell101 out in 10k
.ENDS Lab1_Teeb_Alroshood_R10k

.global gnd

*** TOP LEVEL CELL: DAC5{sch}
XR10k00 b4 net@0 Lab1_Teeb_Alroshood_R10k
XR10k01 net@0 vout Lab1_Teeb_Alroshood_R10k
XR10k03 vout net@15 Lab1_Teeb_Alroshood_R10k
XR10k04 b3 net@3 Lab1_Teeb_Alroshood_R10k
XR10k05 net@3 net@15 Lab1_Teeb_Alroshood_R10k
XR10k06 net@15 net@16 Lab1_Teeb_Alroshood_R10k
XR10k07 b2 net@6 Lab1_Teeb_Alroshood_R10k
XR10k08 net@6 net@16 Lab1_Teeb_Alroshood_R10k
XR10k09 net@16 net@17 Lab1_Teeb_Alroshood_R10k
XR10k10 b1 net@9 Lab1_Teeb_Alroshood_R10k
XR10k11 net@9 net@17 Lab1_Teeb_Alroshood_R10k
XR10k12 net@17 net@18 Lab1_Teeb_Alroshood_R10k
XR10k13 b0 net@12 Lab1_Teeb_Alroshood_R10k
XR10k14 net@12 net@18 Lab1_Teeb_Alroshood_R10k
XR10k15 net@18 net@21 Lab1_Teeb_Alroshood_R10k
XR10k16 net@21 gnd Lab1_Teeb_Alroshood_R10k

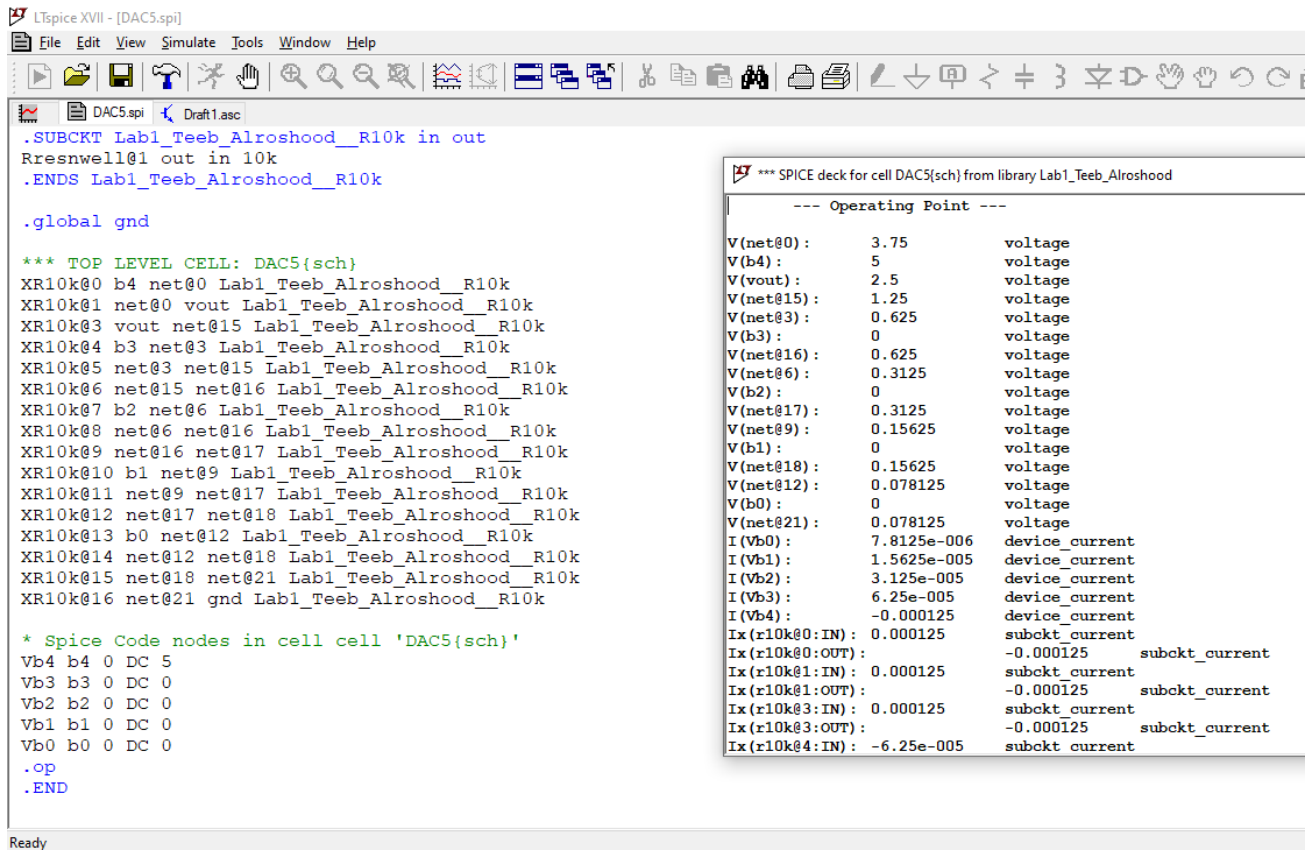
* Spice Code nodes in cell cell 'DAC5{sch}'
Vb4 b4 0 DC 0
Vb3 b3 0 DC 0
Vb2 b2 0 DC 0
Vb1 b1 0 DC 0
Vb0 b0 0 DC 5
.op
.END
```

Ready

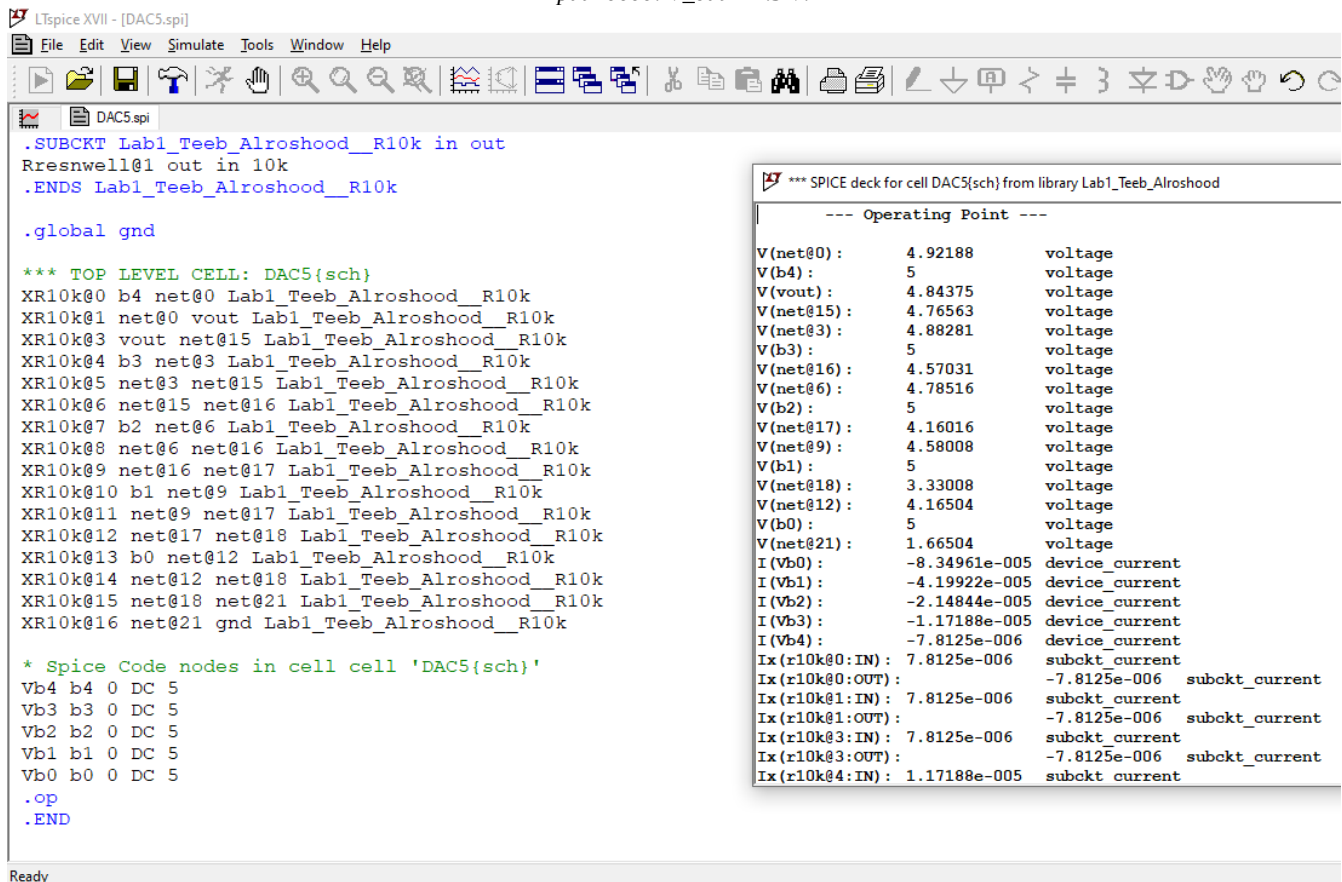
*** SPICE deck for cell DAC5{sch} from library Lab1_Teeb_Alroshood

```
--- Operating Point ---
V(net@0): 0.078125 voltage
V(b4): 0 voltage
V(vout): 0.15625 voltage
V(net@15): 0.234375 voltage
V(net@3): 0.117188 voltage
V(b3): 0 voltage
V(net@16): 0.429688 voltage
V(net@6): 0.214844 voltage
V(b2): 0 voltage
V(net@17): 0.839844 voltage
V(net@9): 0.419922 voltage
V(b1): 0 voltage
V(net@18): 1.66992 voltage
V(net@12): 3.33496 voltage
V(b0): 5 voltage
V(net@21): 0.834961 voltage
I(Vb0): -0.000166504 device_current
I(Vb1): 4.19922e-005 device_current
I(Vb2): 2.14844e-005 device_current
I(Vb3): 1.17188e-005 device_current
I(Vb4): 7.8125e-006 device_current
Ix(r10k00:IN): -7.8125e-006 subckt_current
Ix(r10k00:OUT): 7.8125e-006 subckt_current
Ix(r10k01:IN): -7.8125e-006 subckt_current
Ix(r10k01:OUT): 7.8125e-006 subckt_current
Ix(r10k03:IN): -7.8125e-006 subckt_current
Ix(r10k03:OUT): 7.8125e-006 subckt_current
Ix(r10k04:IN): -1.17188e-005 subckt_current
```

Input 00001: $V_{out} = 0.15625\text{ V}$.



Input 10000: $V_{out} = 2.5$ V.



Input 11111: $V_{out} = 4.84375$ V.

Figure 3. Representative mid-scale and full-scale DC verification.

Delay while driving a 10 pF load

For the delay test, b3 is equal to b2, which is also equal to b1, which is also equal to b0, which is equal to zero; the only difference being that b4 is driven by a 0-5 V pulse and the output is loaded by a capacitance of 10 pF. From there, we can calculate

$$\tau = R_{\text{out}} C_L = (10 \text{ k}\Omega)(10 \text{ pF}) = 100 \text{ ns},$$

$$t_d \approx 0.7 R_{\text{out}} C_L = 70 \text{ ns}.$$

From LTspice, the 50% point of the input step ($V(b4) = 2.5 \text{ V}$) and the 50% point of the output step ($V(\text{vout}) = 1.25 \text{ V}$) gives us the measured delay of

$$t_d = 69.1885 \text{ ns}.$$

This means that if we compare the measured delay to the value obtained from the $0.7RC$, the difference is around 1.16%.

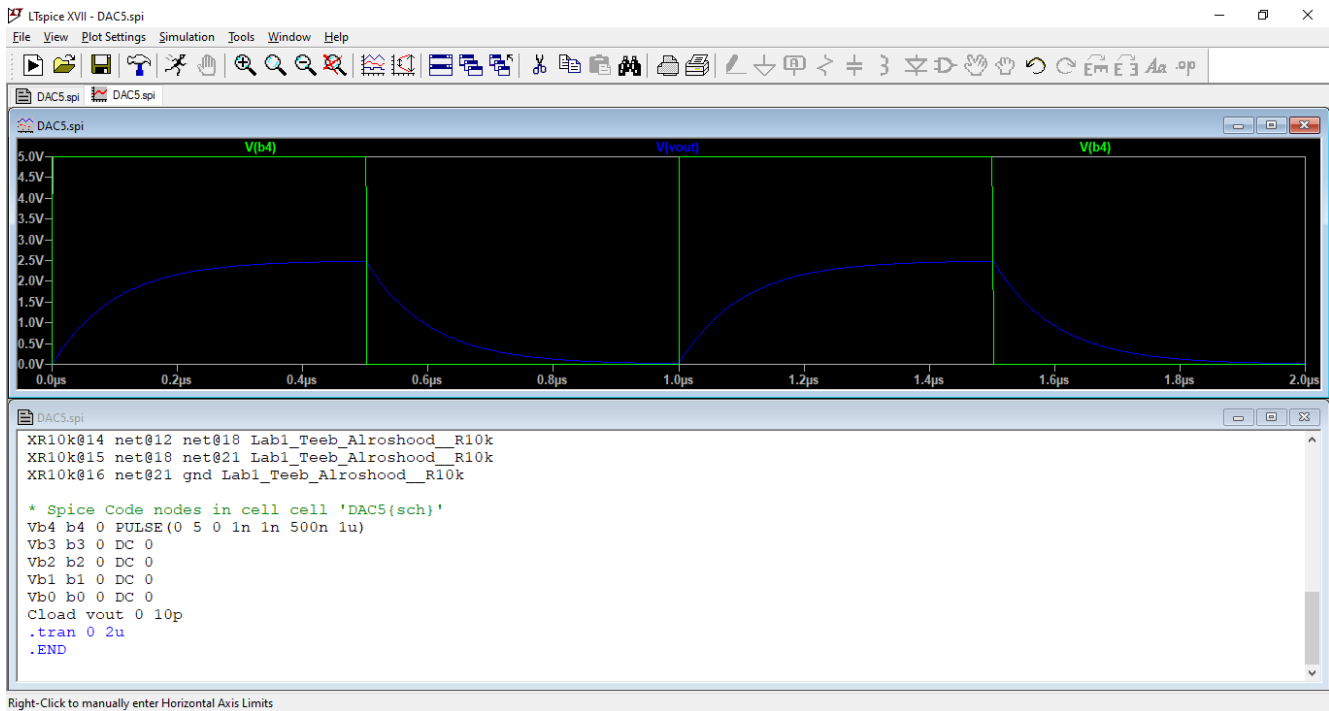


Figure 4. Transient response with a 10 pF load. The green trace is $V(b4)$; the blue trace is $V(\text{vout})$.

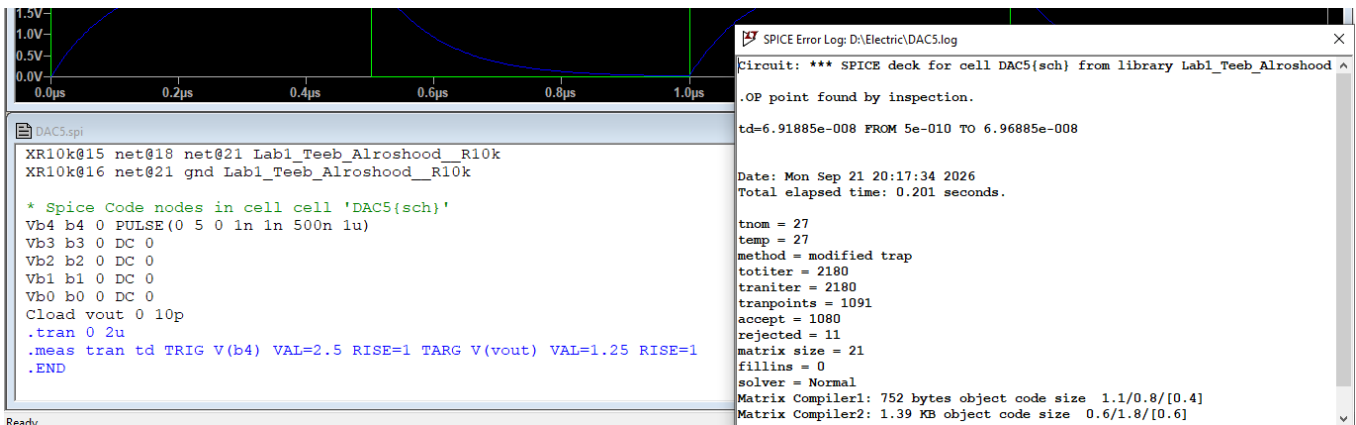


Figure 5. LTspice delay measurement. The reported value is $6.91885 \times 10^{-8} \text{ s} = 69.1885 \text{ ns}$.

10 k Ω load

The output from the DAC can be expressed in terms of its Thevenin equivalent consisting of the ideal DAC voltage and $R_{out} = 10\text{ k}\Omega$ in series. With load resistance of $R_L = 10\text{ k}\Omega$,

$$V_{out}(\text{with load}) = V_{out}(\text{ideally}) * (R_L / (R_{out} + R_L)) = 0.5 V_{out}(\text{ideally}).$$

Thus, for an input of 10,000, the unloaded output would be given by

$$V_{out}(\text{with load}) = 2.5 * (10\text{ k}\Omega / (10\text{ k}\Omega + 10\text{ k}\Omega)) = 1.25\text{ V}.$$

The same output value is produced by LTspice.

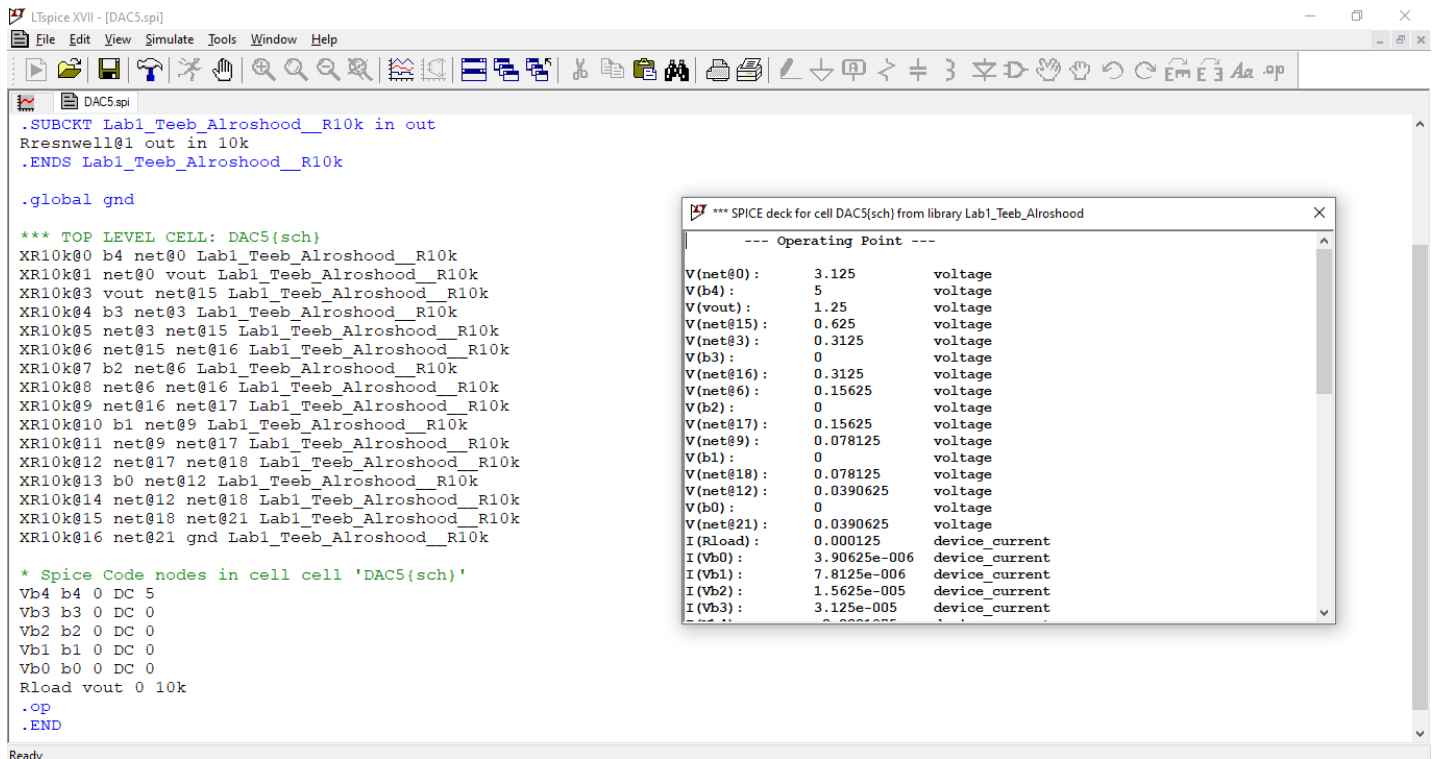


Figure 6. Operating-point simulation for input 10000 with a 10 k Ω load. The loaded output is 1.25 V.

N-well resistor and layout

The Electric course utilized an n-well resistivity of around $800 \Omega/\text{square}$. As a result,

$$L/W = 10,000 / 800 = 12.5$$

The parameters of the Electric cell have the dimensions $W = 15\lambda$ and $L = 187.5\lambda$. Thus, with $\lambda = 0.3 \mu\text{m}$,

$$W = 4.5 \mu\text{m} \text{ and } L = 56.25 \mu\text{m}$$

The complete 5-bit DAC makes use of a total of sixteen $10 \text{ k}\Omega$ n-well resistors, which comprise ten resistors from the five $2R$ input branches, four resistors in the ladder, and two for the termination of the last $2R$, thereby connecting enough resistors in parallel each having the same x-coordinates given different y-coordinates so as to obtain stacked/serpentine arrangement.

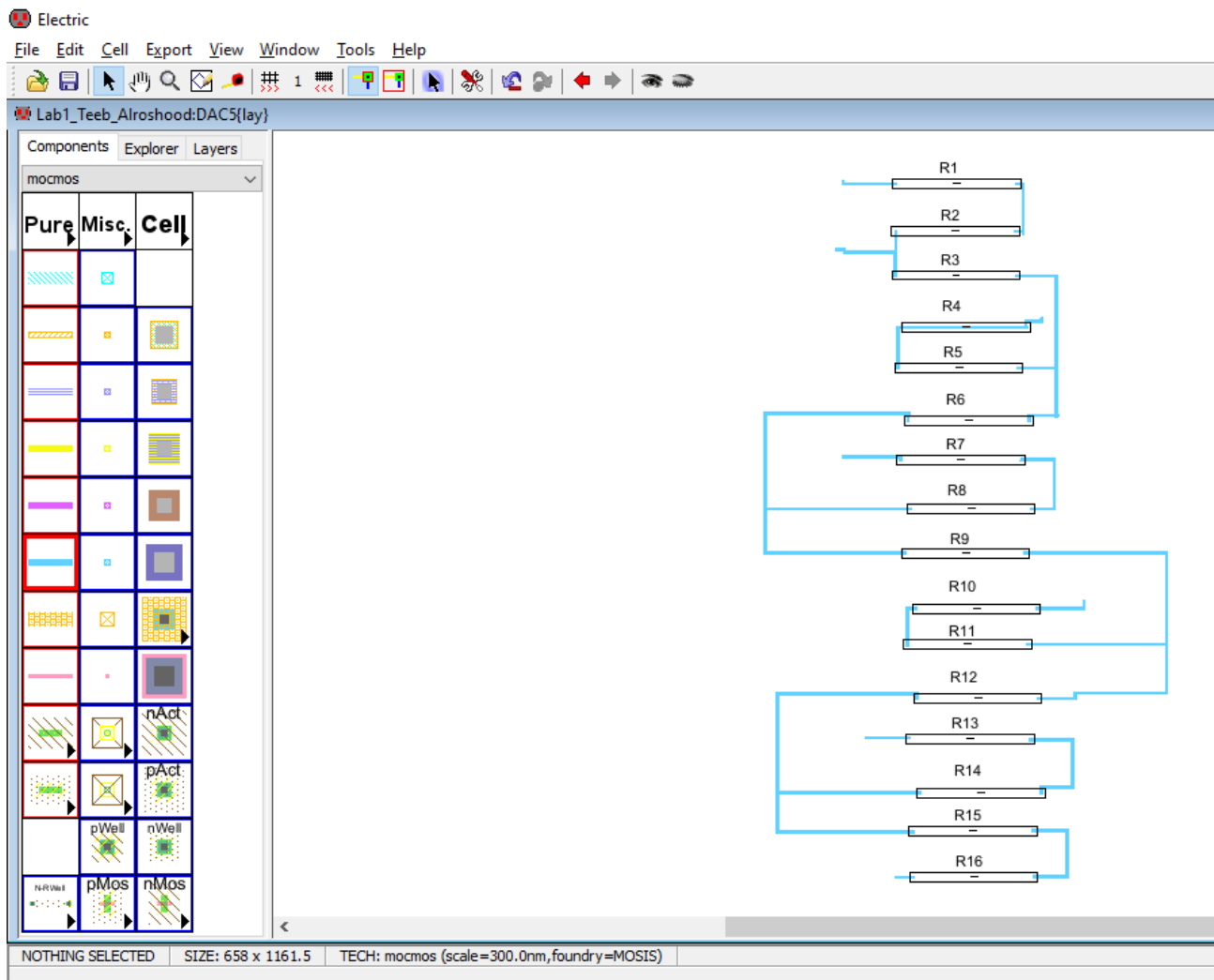


Figure 7. Final Electric DAC layout with sixteen stacked $10 \text{ k}\Omega$ n-well resistor cells and Metal1 routing.

DRC and LVS verification

The final layout meets hierarchical DRC with no errors or warnings. Electric NCC states, in this case, that the schematic and layout exports correspond with each other and that their topologies coincide.

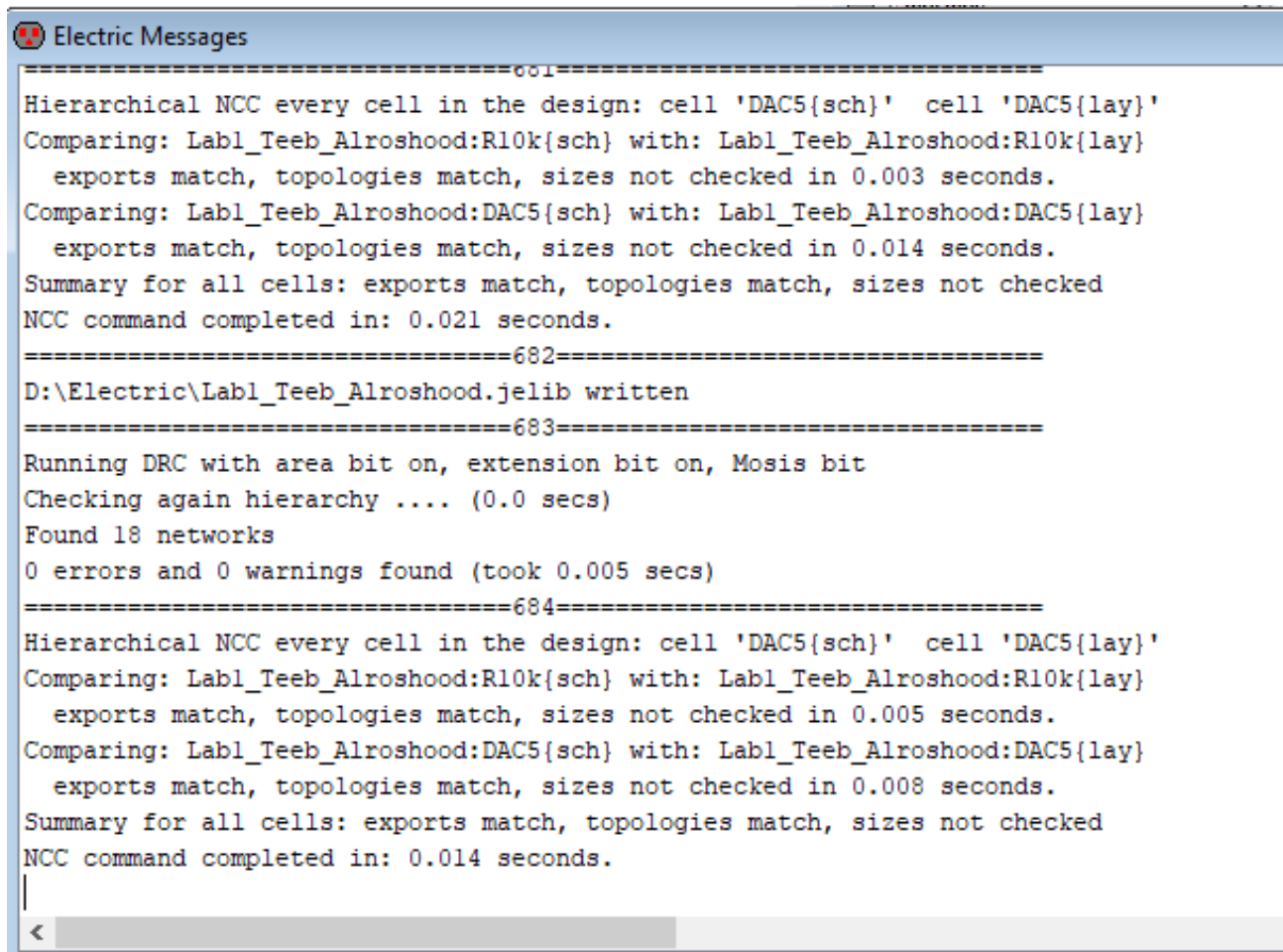
The image shows a screenshot of the 'Electric Messages' window in a software application. The window has a blue header bar with a red error icon and the text 'Electric Messages'. The main area is a text box with a white background and a scrollbar on the right. The text inside the box is a log of verification messages. It starts with a separator line of equals signs. The first section reports a Hierarchical NCC check for cell 'DAC5{sch}' vs 'DAC5{lay}', comparing Lab1_Teeb_Alroshood:R10k{sch} with Lab1_Teeb_Alroshood:R10k{lay}. It states 'exports match, topologies match, sizes not checked in 0.003 seconds.' and 'Comparing: Lab1_Teeb_Alroshood:DAC5{sch} with: Lab1_Teeb_Alroshood:DAC5{lay} exports match, topologies match, sizes not checked in 0.014 seconds.' The summary says 'exports match, topologies match, sizes not checked' and 'NCC command completed in: 0.021 seconds.' This is followed by another separator line. The next section shows 'D:\Electric\Lab1_Teeb_Alroshood.jelib written' followed by another separator line. The third section reports 'Running DRC with area bit on, extension bit on, Mosis bit' and 'Checking again hierarchy (0.0 secs)'. It then states 'Found 18 networks' and '0 errors and 0 warnings found (took 0.005 secs)'. This is followed by a fourth separator line. The final section repeats the Hierarchical NCC check for cell 'DAC5{sch}' vs 'DAC5{lay}', comparing Lab1_Teeb_Alroshood:R10k{sch} with Lab1_Teeb_Alroshood:R10k{lay}. It states 'exports match, topologies match, sizes not checked in 0.005 seconds.' and 'Comparing: Lab1_Teeb_Alroshood:DAC5{sch} with: Lab1_Teeb_Alroshood:DAC5{lay} exports match, topologies match, sizes not checked in 0.008 seconds.' The summary says 'exports match, topologies match, sizes not checked' and 'NCC command completed in: 0.014 seconds.' The text box ends with a vertical bar and a scrollbar at the bottom.

Figure 8. Electric verification messages showing DRC clean and NCC/LVS schematic-layout topology match.

Conclusion

The 5-bit R-2R DAC performs with the predicted LSB of 0.15625 V for a reference voltage of 5 V. The codes used affirm to have conformed to the theoretical relationship between input and output. The measured propagation delay of 10 pF loading is 69.1885 ns which is in close agreement with the estimated value of 70 ns. Connecting a load of 10 k Ω results in output voltage reducing to half the original value which corresponds to the resistance value of the DAC output. The final stacked n-well layout has passed DRC test and is consistent with the schematic according to NCC/LVS test.